pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

Word Embeddings

WEEK 1 · DAY 4

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Words as vectors

Day 3: a column per term

"bond" and "bonds" are two unrelated columns; most of a row is zero.

Today: 100 numbers per word

Words used in similar contexts get rows that point the same way. Every number is filled.

44,558

GloVe words kept by the course

100

numbers per word

0

words with a capital letter

Real documents, described

An 8-K section's model output

Allowed names	Never called
the item the model assigns	a default signal
the model's probability or score in this file	a distress flag or a warning
the filer's item	a credit event or credit view

An indenture's terms

Allowed names	Never called
the indenture states	loose, tight, weak, strong
extracted value, with its quote	covenant-lite or protective
passed the source check	a credit view or recommendation

A word as a vector

44,558

words, float32

100

numbers per word

0

capital letters: GloVe is uncased

```
vectors, words = textkit.load_word_vectors("course7_glove_6b_100d.npz")
position["bond"]                # 1,846: GloVe's frequency order
vectors[position["bond"]][:3]   # 0.8918, 0.3513, 0.1522: nothing on their own
np.linalg.norm(bond)            # 5.8566: compare directions, not lengths
```

Every lookup must lower-case its word first.

Where the vectors come from

The method	an unsupervised learning algorithm ... global word-word co-occurrence statistics
The file	Wikipedia 2014 + Gigaword 5 (6B tokens, 400K vocab, uncased ...): glove.6B.zip
The licence	Public Domain Dedication and License v1.0

Pennington, Socher, Manning: EMNLP 2014, pages 1532 to 1543. word2vec named, not used.

Sources, read 8 October 2026: nlp.stanford.edu/projects/glove; aclanthology.org/D14-1162; arxiv.org/abs/1301.3781

Cosine and nearest words

Pair	Cosine
bond, bonds	0.8374
bond, loan	0.5232
bond, tree	0.1483
default, bankruptcy	0.5639

Word	Its three nearest words
bond	bonds, treasury, benchmark
default	debt, borrower, borrowers
inflation	rate, rates, unemployment
covenant	lutheran, charter, covenants

A nearest word is used in similar contexts in the training text. It is not a meaning.

What is missing

422

of 63,810 tokens have no vector

137

distinct missing tokens

374

capitalised, of 1,840 in 4.12

Missing token	Count
company's	55
defeasance	44
incurrence	27
holder's	12
owner's	12

Lower-cased	Nearest words
notes	note, letters, write
indebtedness	repayments, borrowings, debts
lien	soong, siew, hsieh

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.12.

Arithmetic on vectors, once

Arithmetic	Nearest word	Cosine
bonds - bond + loan	loans	0.8071
paris - france + germany	berlin	0.8810
king - man + woman	queen	0.7834

It always returns some word, right or wrong. A nearest word is not a fact.

Document vectors as features

Item	TF-IDF F1	GloVe mean F1	TF-IDF recall	GloVe mean recall
1.03	0.9263	0.9119	0.8800	0.8800
2.03	0.8528	0.7526	0.8400	0.7300
2.04	0.9109	0.8000	0.9200	0.8000
macro	0.9269	0.8830	0.9278	0.8824

Accuracy: majority item 0.1350, TF-IDF 0.9298, GloVe mean 0.8866.

Fitted on 3,650 sections of 2019 to 2023, scored on 741 of 2024 (validation); 735 of 2025 closed.

Macro F1 0.9269 against 0.8830: the item the model assigns, in this file.

What averaging loses

```
"the trustee may remove the company"
"the company may remove the trustee"   # identical vectors: True
```

The mean of Section 4.12 sits nearest to

this, the, same, only, for; stop words a share of 0.4984

A mean has no word order, and every token weighs the same.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.12.

Excel side by side

Excel side by side (tested in Excel)

`=SUMPRODUCT(A1:A100,B1:B100)/(SQRT(SUMSQ(A1:A100))*SQRT(SUMSQ(B1:B100)))`	0.8374, within 1e-12
`=SQRT(SUMSQ(A1:A100))`	5.8566
`=AVERAGE(D1:H1)`	0.0353: the mean vector, row 1
`=MATCH("Bonds",J1:J5,0)`	2: capitals ignored
`=XLOOKUP("Bonds",J1:J5,K1:K5)`	1.0565
`=EXACT("Bonds",J2)`	FALSE

No word vectors in Excel: the numbers come from GloVe's file. Python's lookup is exact.

Save the day: read the history

1. Start

.gitignore and mlkit, unchanged

2. textkit

three functions, five tests

3. results.csv

three models on validation

```
git log --oneline --stat -3
Log three models of the 8-K items on the 2024 validation sections
textkit day 4: load_word_vectors, nearest_words, document_vectors, 25 tests
Start the day: .gitignore and Course 4's mlkit, unchanged
```

pytest: 25 passed. The newest commit is at the top.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the lookup never lower-cased: 1,458 of 1,840 tokens used, and a missing share of 0.0000.

Code that runs is not the same as code that is right.

What you can do now

- Load word vectors; compare directions with cosine
- Read nearest words as the training text's usage
- Count what the vectors miss in an indenture
- Use mean vectors as features beside TF-IDF
- Cosine and the mean in Excel; lookups and capitals

Week 1, Day 5 FOMC statement tone over time